

SAHAYAK (ककककक)

Autonomous Multi-Engine AI Platform for Disaster Intelligence,
Telecom Silent Zone Prediction & Shelter Supply Redistribution

1. EXECUTIVE SUMMARY

SAHAYAK is a mission-critical AI platform designed for the National Disaster Management and emergency response forces in India. It integrates physical destruction physics, blackout prediction and dynamic multi-commodity shelter supply rebalancing.

2. CORE TRIPLE-ENGINE ARCHITECTURE

- Engine 1: Telecommunication Silent Zone Classifier
 - Predicts cell mast failure, RSSI degradation, and cellular blackout probability.
 - Includes 95% Conformal Confidence Bounds and recall-optimized classification.
- Engine 2: Physical Disaster Impact & Kinetic Destruction Regressor
 - Physics-informed aerodynamic wind stagnation pressure ($q = 0.5 * \rho * v^2$).
 - Hydrodynamic inundation drag and Peukert battery depletion rates (Time-to-Blackout).
- Engine 3: Shelter Resource Dynamics & Multi-Commodity Redistribution Optimizer
 - Solves NDMA / Sphere humanitarian standards for food, drinking water, medical supplies.
 - Generates automated logistics dispatch orders with drone and convoy routing.

System Specifications & REST API Endpoints

3. REST API SERVICE SPECIFICATIONS (FastAPI 2.0)

- POST /predict/integrated : Dual-Engine Joint Cognitive Threat Assessment & CDTI
- POST /predict/silent_zone: Standalone Telecommunication Blackout Classifier
- POST /predict/impact : Standalone Physical Destruction & Hazard Radius Model
- POST /evacuation/route : Dynamic A* Multi-Hazard Potential Field Vector Router
- POST /shelters/analyze : Single Shelter Burn Rate & Buffer Days Horizon Evaluation
- POST /shelters/redistribute : Multi-Commodity Network Supply Redistribution Optimization

4. COMPOUND DISASTER THREAT INDEX (CDTI)

Synthesizes physical destruction and communication blackout risk into unified 4-tier risk index

- Level 1: Low Risk (< 40)
- Level 2: Moderate Danger (40 - 59)
- Level 3: High Severity (60 - 79)
- Level 4: Critical Emergency (≥ 80)

5. GOLDEN HOUR SURVIVAL HALF-LIFE ($\tau_{1/2}$)

Forewarns incident commanders of effective rescue windows before casualty mortality